A3-PS013-TC519

Cyber Threat Intelligence and Incident Correlation Platform

Incident Timeline and Evidence Collection for Small Teams

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Abstract and Problem Description

Abstract

CyberTrace organizes structured MITRE ATT&CK Enterprise threat intelligence for exploration. It combines relational modeling in MySQL with a web prototype for browsing threat entities and relationships. The frontend currently uses structured/mock data and is not yet connected to MySQL through a backend/API.

Problem Description

Threat information is spread across techniques, tactics, software, groups, campaigns, mitigations and detection strategies. A structured, relationship-oriented interface makes it easier to move between these categories.

Objectives

1. Organize threat-intelligence entities in a structured relational model
2. Explore relationships between cybersecurity entities
3. Provide a simple web interface for browsing and searching
4. Lay a foundation for backend integration and expanded investigation workflows



Conceptual illustration of the intended approach; no measured improvement is claimed.

Background

Cyber Threat Intelligence (CTI): Evidence-based knowledge about threats that supports defensive decisions

MITRE ATT&CK: Knowledge base of adversary behaviors and related threat information [1]

Tactics and techniques: Tactics are adversary goals; techniques are how goals are achieved [2]

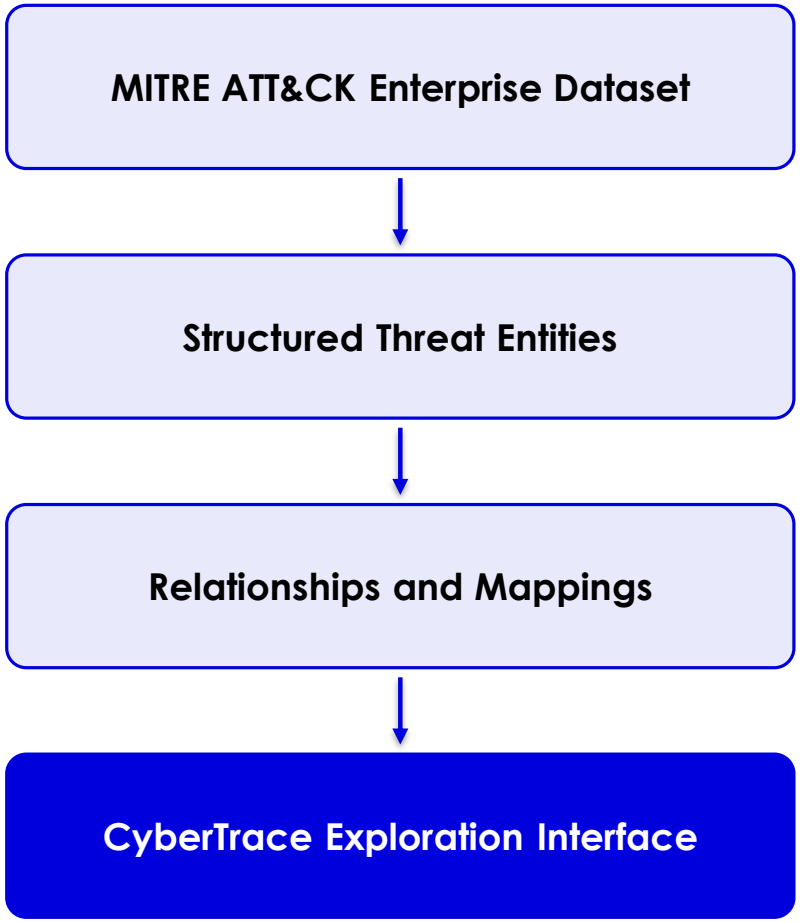
Software and groups: Tools or malware used by adversaries; named threat groups

Campaigns: Sets of intrusion activity described over a period of time

Mitigations: Defensive measures that reduce the impact of techniques

Detection strategies: Guidance on how technique activity can be observed

Relationships: Links connecting the entities above



Sources: MITRE ATT&CK [1][2]

Literature and Standards Review

Concept / Standard	Relevance to CyberTrace
MITRE ATT&CK [1]	Knowledge base of adversary tactics and techniques; the primary source of threat information
STIX 2.1 (OASIS) [3][4]	Standard for representing cyber threat intelligence; ATT&CK data is published in STIX form, informing entity types and relationships
Relational database design [5]	Organizes entities, keys and references in MySQL tables
Normalization and referential integrity [5]	Reduce duplication and keep relationships consistent through foreign keys
Web technologies: HTML5, CSS3, JavaScript, Bootstrap 5.3 [6]	Present structured information in an accessible browser interface

These foundations shape the platform: ATT&CK defines what is modeled, relational design and normalization define how it is stored, and web technologies define how it is explored.

Proposed Solution

1 Threat-intelligence dataset

Structured information from the MITRE ATT&CK Enterprise workbook

2 Relational data layer

MySQL schema designed to represent entities and relationships

3 Web interface prototype

Browser-based exploration using structured/mock demonstration data

4 Future integration layer

Possible backend/API connecting the interface to the database later

FUTURE WORK

Prototype capabilities

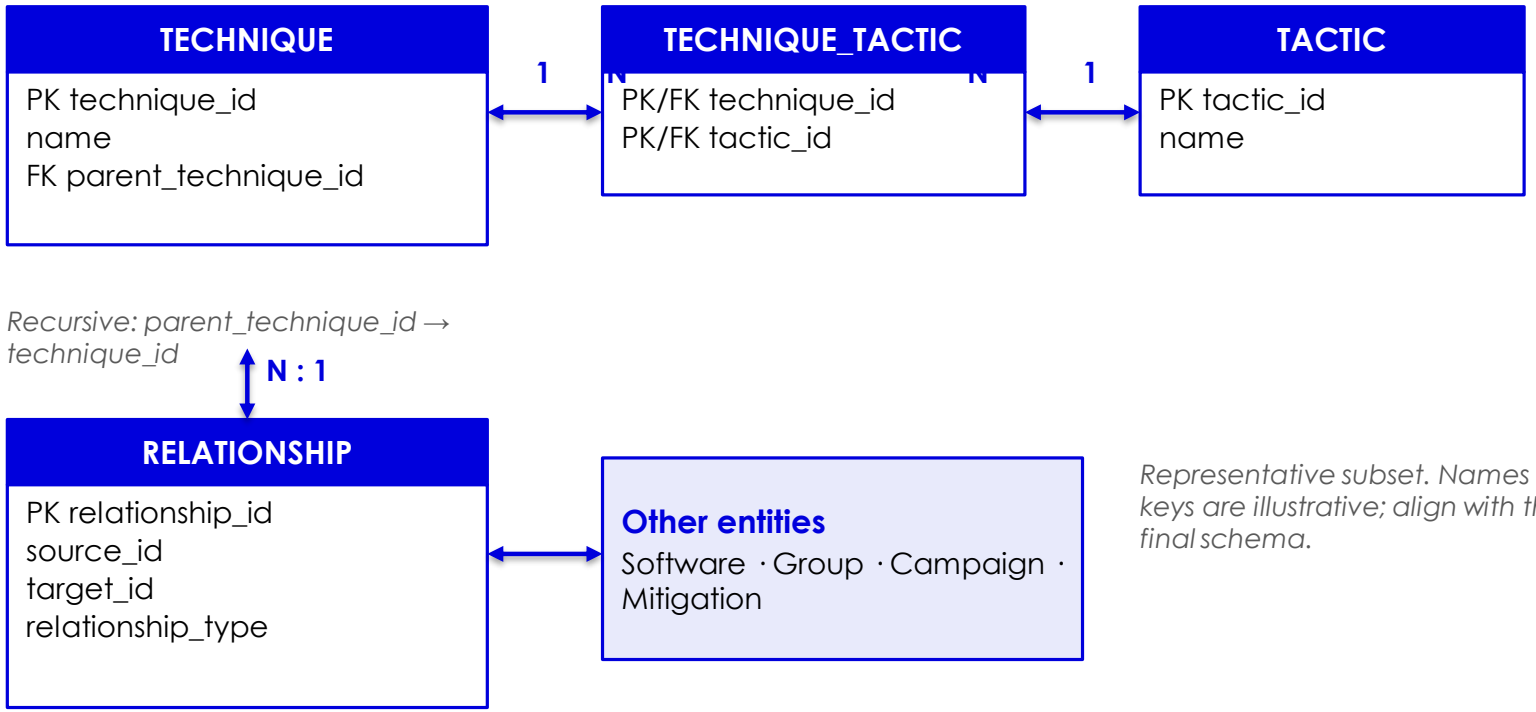
- Browse threat-intelligence categories
- Search and filter, where supported by the prototype
- Inspect individual entity details
- Explore available relationships
- Review illustrative dataset counts and analytics

Implementation status

Current component Future (proposed)

Solid border: current component. Dashed border: proposed future component.

Database Design and Normalization



Representative subset. Names and keys are illustrative; align with the final schema.

Key concepts

- **Primary / foreign keys:** identify and link records
- **One-to-many:** one entity row links to many related rows
- **Many-to-many:** techniques and tactics, via a junction table
- **Recursive:** sub-technique references a parent technique
- **Referential integrity:** foreign keys keep links valid [5]

Normalization

1NF
Atomic values; no repeating lists in one field

2NF
No partial dependency on part of a composite key

3NF
No transitive dependency of non-key attributes

BCNF
Every determinant is a superkey

Why it matters: less duplication, more consistency, fewer update/insert/delete anomalies, clearer relationships. Compliance of each table with a normal form must be verified from the final schema.

Solution Optimization

Design choices in the relational model

(confirm against final schema)

- Primary and foreign keys with integrity constraints
- Junction tables for many-to-many links
- Multi-valued attributes normalized
- Reusable SQL queries; relationships traversed through joins
- Indexes on searched or joined columns, if documented

Recommended future optimizations

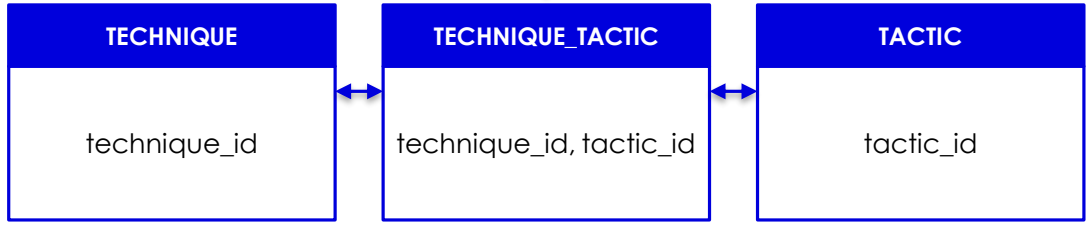
- Index tuning based on real query patterns
- Pagination and filtering through a backend
- Benchmark before claiming any speed gains

Before → After (conceptual illustration)

Denormalized list

technique | tactics (list) | software (list)

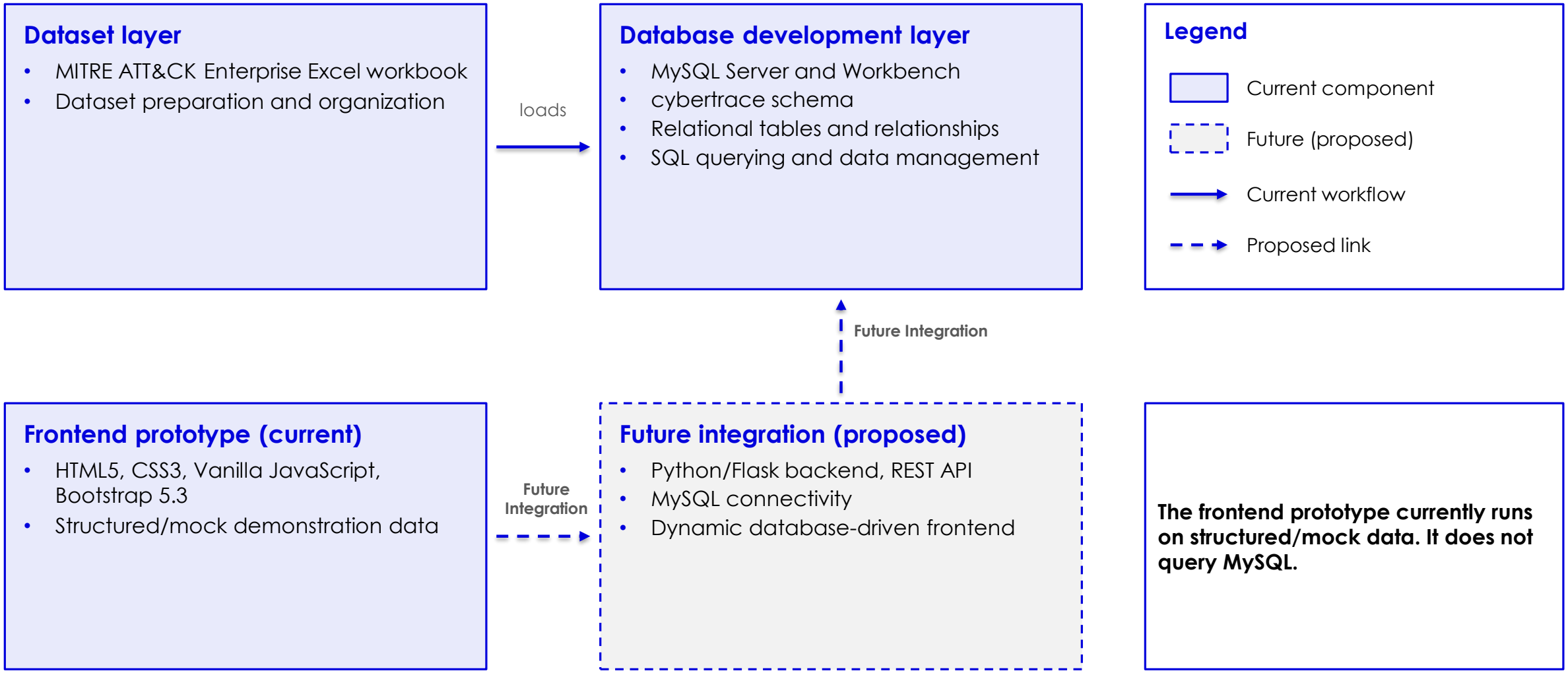
T1059.001 | tactic A, tactic B | tool X, tool Y



Structured entity and junction tables replace repeated lists inside one field.

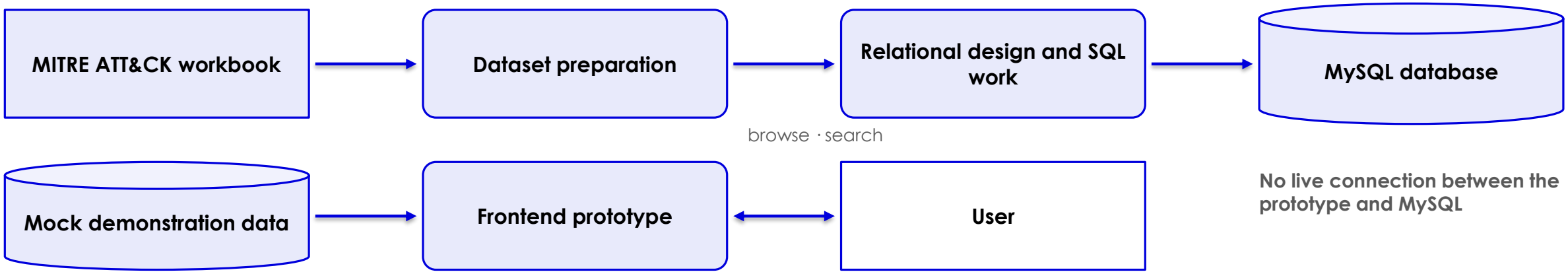
Value: better organization · less duplication · maintainable relationships · clearer retrieval logic · foundation for expansion. No measured speed gains are claimed.

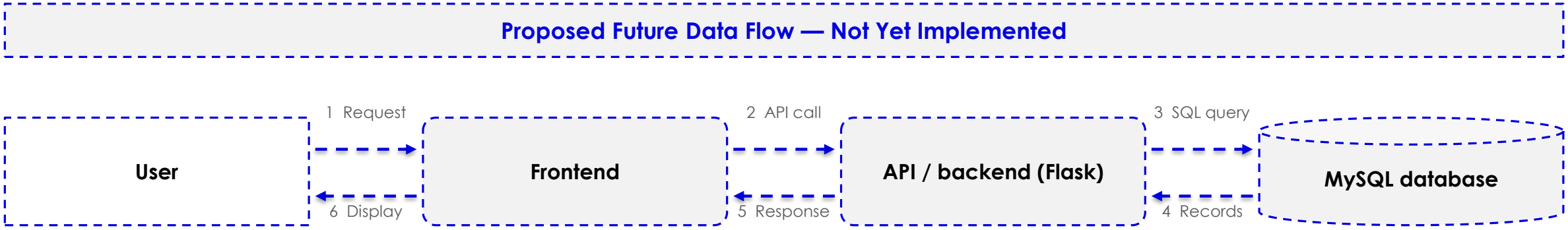
Solution Architecture and Design



Data Flow Diagram

A. Current project workflow

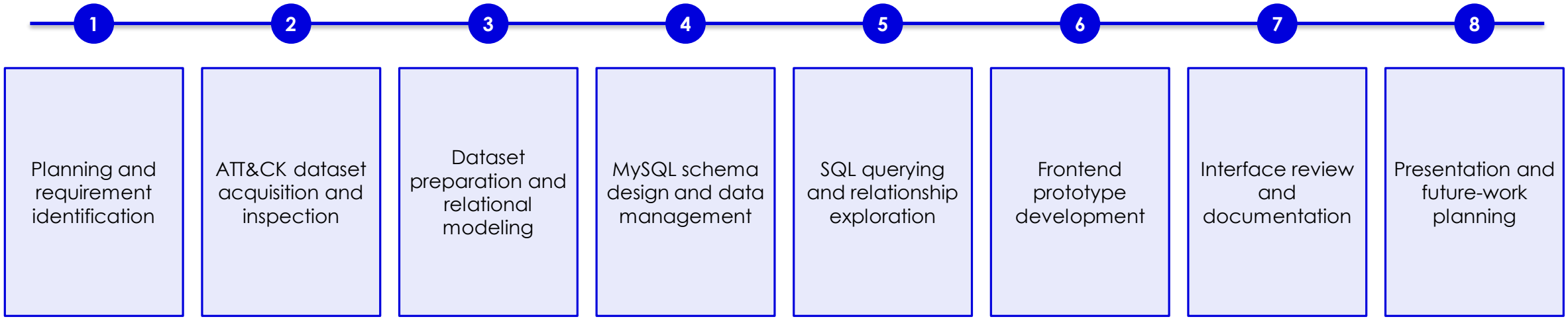




Current component Future (proposed)

Deployment Timeline

Development phases (no dates stated)



Proposed future milestones (not yet started)



Current component Future (proposed)

Application, Results, and Future Scope

Current prototype sections

Dashboard	**Techniques**	**Tactics**	**Software**
Groups	**Campaigns**	**Mitigations**	**Detection Strategies**
Data Components	**Relationships**	**Analytics**	

Dataset/prototype counts (illustrative): 697 techniques · 15 tactics · 825 software · 176 groups · 56 campaigns · 20,628 relationships

Results: verifiable outputs

- Structured dataset inventory
- Relational data model, MySQL schema and SQL work
- Working frontend prototype with navigable sections
- Browsing of available demonstration data

Illustrative use case (browsing scenario)

An analyst investigating suspicious PowerShell activity starts with ATT&CK technique T1059.001 [1] and explores the related techniques, software, groups, mitigations or detection strategies available in the interface. This is not live detection or incident response.

Future scope

- Backend/API integration and dynamic MySQL retrieval
- Data validation and error handling
- User roles and secure authentication
- Import and update workflows for refreshed datasets
- Advanced filtering and relationship visualization
- Optional security-tool integration, testing and usability evaluation

Conclusion and Bibliography

Conclusion

- Organizes MITRE ATT&CK Enterprise information for structured exploration
- Relational modeling represents threat entities and their relationships
- Frontend prototype demonstrates browser-based exploration of threat categories
- Prototype uses structured/mock data; not yet connected to MySQL via a live backend
- Future API integration could enable a database-driven experience

Bibliography

[1] MITRE ATT&CK. attack.mitre.org

[2] MITRE ATT&CK Enterprise Matrix. attack.mitre.org/matrices/enterprise/

[3] MITRE ATT&CK STIX Data (attack-stix-data). github.com/mitre-attack/attack-stix-data

[4] OASIS. STIX Version 2.1. docs.oasis-open.org/cti/stix/v2.1/stix-v2.1.html

[5] Oracle. MySQL 8.4 Reference Manual. dev.mysql.com/doc/refman/8.4/en/

[6] Bootstrap 5.3 Documentation. getbootstrap.com/docs/5.3/

"CyberTrace provides a foundation for structured threat-intelligence exploration and future incident-investigation workflows."

Thank You